

Show all work. Indicate if values determined via shock table or code, if not otherwise obvious from the work. Submit any code used. If an LLM is used, attach a separate chat thread for each problem.

ME 561 students are required to typeset assignments using L^AT_EX. **ME 461** students receive a 2% bonus for typesetting assignments using L^AT_EX.

1. (24 pts) A jet plane is flying at an altitude of 10 km, where the outside conditions are 26.5 kPa and 223.3 K, with a cabin pressure of 101 kPa and temperature of 20 °C. Suddenly, a hole forms in the fuselage, with a hole 2 cm in diameter. Assume that the temperature within the cabin remains constant and that the flow through the hole behaves as that through a converging nozzle with an exit diameter of 2.0 cm. Take the cabin volume to be 100 m³. Calculate the time for the cabin pressure to decrease to one-half the initial value. (*Hint*: this is an unsteady problem.)
2. (16 pts) A supersonic nozzle possessing an area ratio (exit to throat) of 3.0 is supplied from a large reservoir and is allowed to exhaust to atmospheric pressure (101 kPa).
 - (a) Determine the range of reservoir pressures over which a normal shock will appear in the nozzle.
 - (b) For what value of reservoir pressure will the nozzle be perfectly expanded, with supersonic flow at the exit plane?
 - (c) Find the minimum reservoir pressure to produce sonic flow at the nozzle throat.

Assume isentropic flow except for shocks, with $\gamma = 1.4$.

3. (16 pts) The formula for the Mach number downstream of a normal shock is

$$M_2^2 = \frac{M_1^2 + \frac{2}{\gamma-1}}{\frac{2\gamma M_1^2}{\gamma-1} - 1}.$$

Prove that the downstream Mach number $M_2 < 1$, that is, that the flow is always subsonic after passing through the shock.

4. **ME 561 ONLY** (24 pts) A closed tube contains nitrogen at 20°C and a pressure of $1 \times 10^4 \text{ N/m}^2$, as shown in Figure 1. A shock wave progresses through the tube at a speed of 380 m/s .

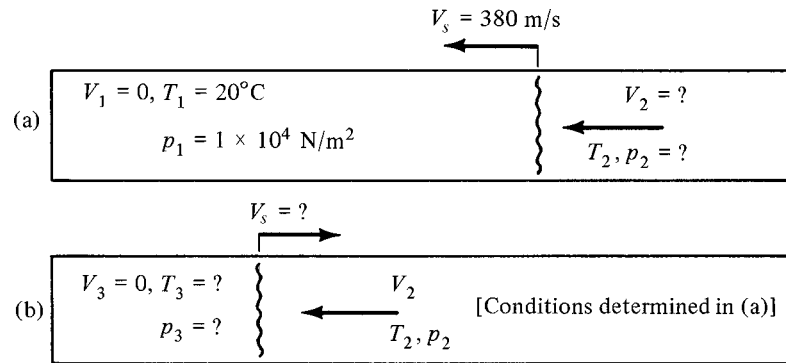
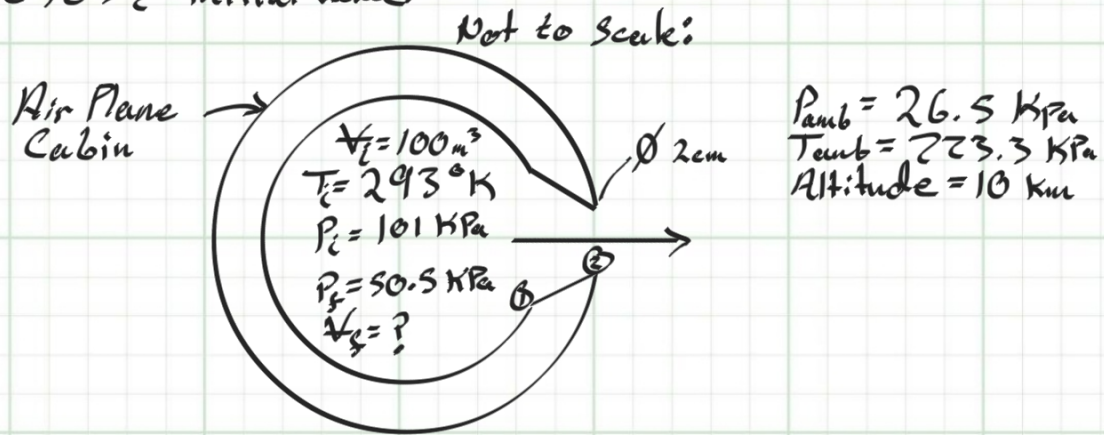


Figure 1: Closed tube with moving shock

- Calculate the conditions that exist immediately after the shock wave passes a given point. (The fact that this is inside a tube should not bother you, as it is merely a normal shock moving into a gas at rest.)
- When the shock wave hits the end wall, it is reflected back. What are the temperature and pressure of the gas between the wall and the reflected shock? At what speed is the reflected shock traveling? (This is just like the sudden closing of a valve in a duct.)

Problem 1

A jet plane is flying @ an altitude of 10 km. The surrounding conditions are 26.5 kPa and 223.3 °K, w/ a Cabin Pressure of 101 kPa and Temp of 20 °C. A hole forms in the fuselage w/ a 2 cm ϕ . Assume: $T_i = c$, and Air flow through the hole acts like converging Nozzle w/ exit ϕ of 2 cm. Cabin Volume is 100 m³. Calculate the time it takes to decrease the Cabin Pressure to 1/2 initial value.



Assuming

Choked flow @ throat $\Rightarrow M = \frac{V}{a} = 1 \quad a = \sqrt{\gamma g_c R T}$

Isentropic (ie No losses) $\Rightarrow V = a$

$$\dot{m} = \rho A_t V \quad \text{is @ } M=1 \quad \frac{P_t}{P_c} = \frac{P_t}{P_i} = \frac{P_t}{P_c}$$

$$\therefore \frac{P_t}{P} @ M=1 = \left(1 + \left(\frac{\gamma-1}{2}\right) M^2\right)^{\frac{\gamma}{\gamma-1}} \xrightarrow{0.2} \frac{P_t}{P} = (1.89) \Rightarrow P_t = \frac{P_c}{1.89}$$

$\xrightarrow{\pi \times 10^{-4} \text{ m}^2} \therefore \frac{\pi (\phi)^2}{4} = \frac{\pi (2 \text{ cm})^2}{4} = \pi \text{ cm}^2$

$$\Rightarrow \dot{m}_t = \frac{P_t A_t a_t}{R T_t} \Rightarrow \frac{P_c A \sqrt{\gamma g_c R T_t}}{1.89 R T_t}$$

Some constant

$$C = \frac{A \sqrt{\gamma g_c R T_t}}{1.89 R T_t} \quad \dot{m}_t = C P_c$$

$$T_t = T_t \left(1 + \frac{\gamma-1}{2} M^2\right)^{-1} \quad \left| \quad T_t = T_1 \quad \therefore \text{isentropic reference Relation} \right.$$

$\hookrightarrow 293 \text{ K} \quad \hookrightarrow 0.2 \quad \hookrightarrow 1$

$$T_t = 244.16 \text{ K}$$

$$\frac{dN_{cm}}{dt} = \frac{\delta}{\delta t} \int_{CV} \rho \vec{v} dV + \int_{CS} \rho \vec{v} (\vec{v} \cdot \hat{n}) dA$$

$$\Rightarrow \frac{dM}{dt} = \frac{\delta M_{cv}}{\delta t} + \dot{m}_2 \quad \because \text{Control mass doesn't change } m_i = m_f \Rightarrow \Delta m = 0$$

$$\Rightarrow \frac{\delta M_{cv}}{\delta t} = -\dot{m}_2 = -C P_c$$

$$M_{cv} = \rho V = \frac{P_c V_c}{RT_c}$$

$$\Rightarrow \frac{\delta}{\delta t} \left\{ \frac{P_c V}{RT_c} \right\} = -C P_c \quad \frac{V_c}{RT_c} = K \quad \begin{matrix} \nearrow 100 \text{ m}^3 \\ \searrow 293 \text{ K} \end{matrix} \quad \text{some constant}$$

$$\Rightarrow K \delta P_c = -C P_c dt$$

$$\Rightarrow \frac{-K}{C} \int_{P_i}^{P_f} \frac{\delta P_c}{P_c} = \int_0^{t_f} dt$$

$$\Rightarrow \frac{-K}{C} \ln \left(\frac{P_f}{P_i} \right) = t_f$$

$$K = \frac{100}{(287)(293)} = 0.001189$$

$$C = \frac{\pi \times 10^{-4} \sqrt{1.4 (287) (244.16)}}{1.89 (287) (244.16)} = 7.4217 \times 10^{-7}$$

$$\frac{-K}{C} = -1600.577$$

$$\therefore t_f = -1600.577 \ln \left(\frac{1}{2} \right) = 1109.4 \text{ s}$$

$$\approx 18.5 \text{ min}$$

The time it takes for the Pressure in the Cabin to decrease by half is $\approx 1110 \text{ s}$ or 18.5 min

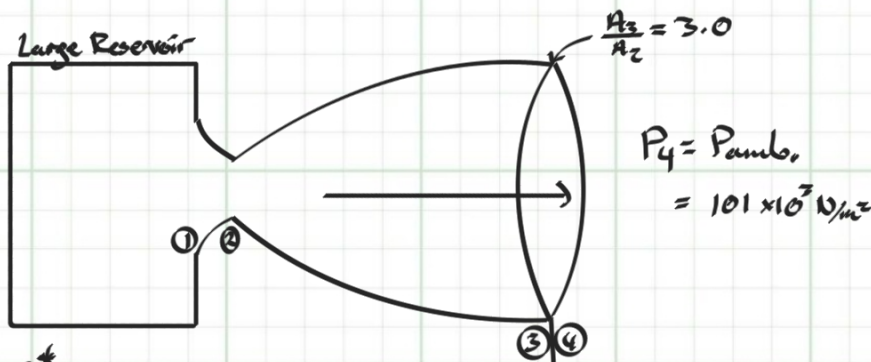
Problem 2

A supersonic Nozzle w/ $A_2/A_t = 3.0$ is supplied from a large reservoir and is allowed to exhaust to 1 atm (101 kPa).

Determine

- The Range of Reservoir Pressures which a Normal Shock will appear in the Nozzle
- What Reservoir Pressure will the Nozzle be perfectly expanded w/ supersonic flow at the exit?
- Minimum Reservoir Pressure to produce sonic flow @ the throat.

Assume: Ideal gas, Isentropic flow except across the shock. $\gamma = 1.4$



$$\therefore A_2 = A^*$$

M	P/P_t	A/A^*	$PA/P_t A^*$
0.20	0.9725	3.0	2.9175
2.64	0.04711	3.0	0.14133

if Subsonic @ exit $\left(\frac{P_t}{P}\right) P_4 = P_{t4} = (0.9725)(101 \text{ kPa}) = 103.4 \text{ kPa}$

Supersonic @ exit $\Rightarrow \left(\frac{P_t}{P}\right) P_4 = P_{t4} = (0.04711)(101 \text{ kPa}) = 2.143.9 \text{ kPa}$
 $M = 2.64$

if shock @ exit $\Rightarrow 2^{nd}$

M_3	M_4	P_4/P_3	$\Delta V/a_1$	P_{t3}/P_{t4}	P_{t4}/P_3
2.64	0.50	7.96	1.88434	0.44522	4.45064

$$\left(\frac{P_3}{P_{t4}}\right) P_{t4} = P_3 = (4.45064)(2143.9 \text{ kPa}) = 226.9 \text{ kPa}$$

$$\left(\frac{P_4}{P_3}\right) P_3 = (7.96)(226.9 \text{ kPa}) \Rightarrow P_4 = 1806 \text{ kPa}$$

$$\left(\frac{P_{t2}}{P_{t4}}\right) P_{t4} = P_{t3} = (0.44522)^{-1} (2143.9 \text{ kPa})$$

$$= 4815.4 \text{ kPa}$$

$\therefore$ isentropic $P_1 = P_{t3} = 4815.4 \text{ kPa}$

Shock @ throat $\Rightarrow$ 1st Crit

M_2	M_3	P_3/P_2	T_3/T_2	$\Delta V/a_1$	P_{t3}/P_{t2}	P_{t3}/P_2
1.00	1.00	1.00	1.00	0	1.00	1.893

$$\left(\frac{P_2}{P_{t3}}\right) (P_{t3}) = P_2 = P_3 = (1.893)^{-1} (4815.4 \text{ kPa})$$

$$= 2543.9 \text{ kPa}$$

$P_2 = P_3 = 2.5 \text{ MPa} = \text{Receiver Pressure max}$

$\Rightarrow$ Minimum Reservoir Pressure

Maximum Reservoir Pressure Required to induce shock @ exit

$$P_1 = 4.8 \text{ MPa}$$

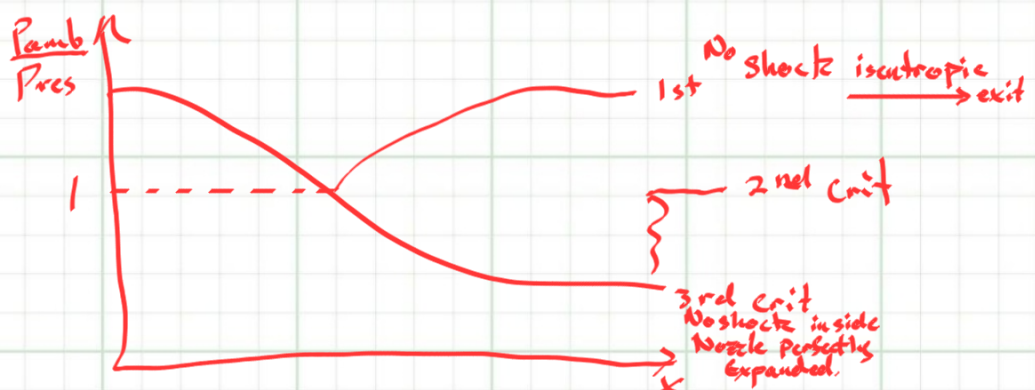
Minimum Reservoir Pressure Required to induce shock @ throat

$$P_1 = 2.5 \text{ MPa}$$

Wrong, discussed in office hours, know what I

$\rightarrow$ Need to do for corrections

relate pressures to 1st crit 2nd crit + 3rd crit



Problem 3

The general Formula for the downstream Mach of a Normal Shock:

$$M_2^2 = \frac{M_1^2 + \frac{2}{\gamma-1}}{\frac{2\gamma M_1^2}{\gamma-1} - 1}$$

Prove that $M_2 < 1$ i.e. (flow is always subsonic after transitting the Shock.)

$$\begin{aligned} M_2^2 &= \frac{M_1^2(\gamma-1) + 2}{\frac{2\gamma M_1^2(\gamma-1)}{\gamma-1}} = \frac{M_1^2(\gamma-1) + 2}{2\gamma M_1^2 - \gamma - 1} = \frac{(M_1^2(\gamma-1) + 2)(2\gamma M_1^2 - \gamma - 1)}{(2\gamma M_1^2 - \gamma - 1)^2} \\ &= \frac{(M_1^2\gamma - M_1^2 + 2)(2\gamma M_1^2 - \gamma - 1)}{4\gamma^2 M_1^4 - 4\gamma M_1^2(\gamma-1) + (\gamma-1)(\gamma-1)} \\ &= \frac{2\gamma^2 M_1^4 - M_1^2\gamma^2 + M_1^2\gamma - 2\gamma M_1^4 + M_1^2\gamma - M_1^2 + 4\gamma M_1^2 - 2\gamma + 2}{4\gamma^2 M_1^4 - 4\gamma M_1^2\gamma + 4\gamma M_1^2 + \gamma^2 - 2\gamma + 1} \\ &= \frac{2\gamma^2 M_1^4 - \gamma^2 M_1^2 + 6\gamma M_1^2 - 2\gamma M_1^4 - M_1^2 - 2\gamma + 2}{4\gamma M_1^2(\gamma M_1^2 - \gamma + 1) + \gamma^2 - 2\gamma + 1} \\ &= \frac{2\gamma M_1^4(\gamma-1) - M_1^2(\gamma^2 - 6\gamma + 1) - 2(\gamma-1)}{4\gamma M_1^2(\gamma M_1^2 - \gamma + 1) + \gamma^2 - 2\gamma + 1} \end{aligned}$$

∴ @ $M_1 = 1$

$$\frac{2\gamma(\gamma-1) - (\gamma^2 - 6\gamma + 1) - 2(\gamma-1)}{4\gamma(\gamma - \gamma + 1) + \gamma^2 - 2\gamma + 1}$$

$$\Rightarrow \frac{3\gamma^2 - 2\gamma + 6\gamma - 2\gamma + 1}{\gamma^2 + 2\gamma + 1} \Rightarrow \frac{3\gamma^2 + 2\gamma + 1}{\gamma^2 + 2\gamma + 1} \Rightarrow \text{ratio of } 1.569$$

$$\Rightarrow \text{as } M_1 \rightarrow \infty \text{ H.O.T } M_2^2 = \frac{2\gamma^2 M_1^4}{4\gamma^2 M_1^4} \Rightarrow M_2^2 \propto \frac{1}{2}$$

∴ Because Higher Order terms M_2^2 is $\propto \frac{1}{2}$ of M_1

Wrong

$$\frac{\mu_1^2 + \frac{2}{\gamma-1}}{2\gamma\mu_1^2 - 1}$$

$$\frac{1 + \frac{2}{\mu_1^2(\gamma-1)}}{\frac{2\gamma}{\gamma-1} - \frac{1}{\mu_1^2}}$$

$\rightarrow 0 \because \text{Lim } \mu_1^2 \rightarrow \infty$
 $\rightarrow 0$

$$\Rightarrow \frac{\gamma-1}{2\gamma} \Rightarrow \mu_2^2 < 1$$

Weed Plot $\mu_2^2 = \frac{\mu_1^2 + \frac{2}{\gamma-1}}{2\gamma\mu_1^2 - 1}$

and Vary μ_1 as μ_1 increase

then Vary γ as γ increases. Where $\mu_1 = 1$

since γ for monatomic gases caps out @ $\frac{5}{3}$

Set $1 \leq \gamma \leq \frac{5}{3}$ to show effect on μ_2^2